

D3A

$$a_n = \left(\frac{n^3 - 2}{n^3 + 1} \right)^{n^3}$$

$$\lim_{n \rightarrow +\infty} \left(\frac{n^3 - 2}{n^3 + 1} \right)^{n^3} = (n^3 - 2)^{n^3} \cdot \left[(n^3 + 1)^{n^3} \right]^{-1} = e^{-2} \cdot e^{-1} = e^{-3}$$

D3B

$$a_n = \left(\frac{n^2 - 3}{n^2 + 2} \right)^{n^2}$$

$$\lim_{n \rightarrow +\infty} \left(\frac{n^2 - 3}{n^2 + 2} \right)^{n^2} = (n^2 - 3)^{n^2} \cdot \left[(n^2 + 2)^{n^2} \right]^{-1} = e^{-3} \cdot e^{-2} = e^{-5}$$

D3C

$$a_n = \left(\frac{n^4 + 4}{n^4 + 2} \right)^{n^4} = (n^4 + 4)^{n^4} \cdot \left[(n^4 + 2)^{n^4} \right]^{-1} = e^4 \cdot e^{-6} = e^2$$

D4A

$$\lim_{n \rightarrow +\infty} \frac{n^2 - 3n(\ln n)^{10} - n\sqrt{n^3+1}}{2n^2 + e^{1/n} - n\sqrt{n}} = \frac{n^2 - 3n(\ln n)^{10} - n(n^{3/2+1})}{2n^2 + e^{1/n} - n \cdot n^{1/2}} = \frac{n^2 - 3n(\ln n)^{10} - n^{5/2} - n}{2n^2 + e^{1/n} - n^{3/2}} \sim \frac{-n^{5/2}}{2n^2} = -\infty$$

$$\ln(n) \ll x^x \ll e^x \ll n! \ll n^n$$

D4B

$$\lim_{n \rightarrow +\infty} \frac{n^3 - 4n^2(\ln n)^{11} - e^{1/n}}{2n^3 - n^2\sqrt{n^3+1} - n\sqrt{n}} = \frac{n^3 - 4n^2(\ln n)^{11} - e^{1/n}}{2n^3 - n^2(n^{3/2+1}) - n^{3/2}} = \frac{n^3 - 4n^2(\ln n)^{11} - e^{1/n}}{2n^3 - n^{7/2} - n^2 - n^{3/2}} \sim \frac{n^3}{-n^{7/2}} = 0$$

D4C

$$\lim_{n \rightarrow +\infty} \frac{-n^3 - 3n^2(\ln n)^7 + n^2(n^{3/2} + 7^{1/2})}{4n^3 - n^{5/2} + e^{1/n}} = \frac{-n^3 - 3n^2(\ln n)^7 + n^{7/2} + \sqrt{7}n^2}{4n^3 - n^{5/2} + e^{1/n}} \sim \frac{n^{7/2}}{4n^3} = +\infty$$

D5A

$$y = 4e^{\sqrt{x}}$$

$$y' = 4e^{\sqrt{x}} \cdot \frac{1}{2\sqrt{x}} = \frac{2e^{\sqrt{x}}}{\sqrt{x}}$$

$$f'(4) = \frac{2e^{\sqrt{4}}}{\sqrt{4}} = \frac{2e^2}{2} = e^2$$

D5B

$$f(x) = \ln(1 + \sqrt{x})$$

$$f'(x) = \frac{1}{1 + \sqrt{x}} \cdot \frac{1}{2\sqrt{x}} = \frac{1}{2\sqrt{x}(1 + \sqrt{x})} = \frac{1}{2\sqrt{x} + 2x}$$

$$f'(4) = \frac{1}{2 \cdot 2 + 2 \cdot 4} = \frac{1}{12}$$

D5c

$$f(x) = \sqrt{x^2 + 9}$$

$$f'(x) = \frac{1}{2\sqrt{x^2 + 9}} \cdot 2x = \frac{2x}{2\sqrt{x^2 + 9}}$$

$$f'(4) = \frac{8}{2\sqrt{16 + 9}} = \frac{4}{5}$$

D6A

$$y = \sin(x^2)$$

$$\lim_{x \rightarrow 0} \frac{\sin(x^2)}{x} = \frac{\frac{\sin(x^2)}{x^2} \cdot x^2}{x} = \frac{x^2}{x} = x = 0$$

D6B

$$f(x) = e^{x^3} - 1$$

$$\lim_{x \rightarrow 0} \frac{e^{x^3} - 1}{x^3} = 1, \lim_{x \rightarrow 0} \frac{e^{x^3} - 1}{x^4} \stackrel{\text{D6c}}{=} \lim_{x \rightarrow 0} \frac{e^{x^3}}{x^4} = \lim_{x \rightarrow 0} \frac{e^{x^3}}{\ln(1 + x^4)} \cdot \frac{1}{x} = 1 \cdot +\infty = +\infty$$

$$\lim_{x \rightarrow 0} \frac{e^{x^3} - 1}{x^2} = \frac{e^{x^3} - 1}{x^3} \cdot x^3 \cdot \frac{1}{x^2} \stackrel{\text{D6c}}{=} \lim_{x \rightarrow 0} \frac{\ln(1 + x^4)}{x^3} = \frac{\ln(1 + x^4)}{x^4} \cdot x^4 \cdot \frac{1}{x^2} = x = 0$$

D7A

$$f(x) = -x^3 + x - 2$$

$$f'(x) = -3x^2 + 1$$

$$-3x^2 + 1 = 0 \Rightarrow -3x^2 = -1 \Rightarrow 3x^2 = 1 \Rightarrow x^2 = \frac{1}{3} \Rightarrow x = \pm \frac{1}{\sqrt{3}}$$

$$f(1/\sqrt{3}) = \left(-\frac{1}{\sqrt{3}}\right)^3 + \frac{1}{\sqrt{3}} - 2 = -\frac{1}{3\sqrt{3}} + \frac{1}{\sqrt{3}} - 2 = \frac{-1 + 3 - 6\sqrt{3}}{3\sqrt{3}} = \frac{2 - 6\sqrt{3}}{3\sqrt{3}} = \frac{2}{3\sqrt{3}} - 2$$

$$f(2) = -8 + 2 - 2 = -8$$

$$f(1) = -1 + 1 - 2 = -3 + 1 = -2$$

D7B

$$f(x) = -x^3 + 2x - 1$$

$$f(1) = -1 + 2 - 1 = 0$$

$$f(2) = -8 + 4 - 1 = -5$$

$$f'(x) = -3x^2 + 2$$

$$-3x^2 + 2 = 0 \Rightarrow 3x^2 = 2 \Rightarrow x^2 = 2/3 \Rightarrow x = \pm \sqrt{2/3}$$

$$f(\sqrt{2/3}) = -\frac{2\sqrt{2}}{3\sqrt{3}} + \frac{2\sqrt{2}}{\sqrt{3}} - 1 = \frac{-2\sqrt{2} + 6\sqrt{2} - 3\sqrt{3}}{3\sqrt{3}} = \frac{4\sqrt{2} - 3\sqrt{3}}{3\sqrt{3}} = \frac{4\sqrt{2}}{3\sqrt{3}} - 1$$

D7C

$$f(x) = -2x^3 + x + 1$$

$$f(1) = -2 + 1 + 1 = 0$$

$$f(2) = -16 + 2 + 1 = -13$$

$$f'(x) = -6x^2 + 1$$

$$-6x^2 + 1 = 0 \Rightarrow 6x^2 = 1 \Rightarrow x = \pm \sqrt{1/6}$$

D8A

$$\int_0^2 t e^t = \left[t e^t - e^t \right]_0^2 = (2e^2 - e^2) - (0 - 1) = e^2 + 1$$

$$\int t e^t = t e^t - \int e^t = t e^t - e^t$$

F8 F6 f6

D8B

$$\int_{-1}^1 t e^t = [t e^t - e^t]_{-1}^1 = (e^1 \cdot e^1) - (-e^{-1} - e^{-1}) = 2e^{-1}$$

$$\int_{\text{FG}} t e^t = \underset{\text{FG}}{t e^t} - \int_{\text{FG}} e^t = \underset{\text{FG}}{t e^t} - e^t$$

D8e

$$\int_{-2}^0 t e^t = [t e^t - e^t]_{-2}^0 = (0 - 1) - (-2e^{-2} - e^{-2}) = -1 + 3e^{-2}$$

$$\int_{\text{FG}} t e^t = \underset{\text{FG}}{t e^t} - \int_{\text{FG}} e^t = \underset{\text{FG}}{t e^t} - e^t$$

D2A

$$\sum (2x-1)^n$$

Condizione
convergenza

$$|2x-1| < 1 \Rightarrow -1 < 2x-1 < 1 \Rightarrow 0 < 2x < 2 \Rightarrow 0 < x < 1$$

D2B

$$\sum (2x+1)^n$$

Condizione
convergenza

$$|2x+1| < 1 \Rightarrow -1 < 2x+1 < 1 \Rightarrow -2 < 2x < 0 \Rightarrow -1 < x < 0$$

D2e

$$\sum (1+3x)^n$$

Condizione
convergenza

$$|1+3x| < 1 \Rightarrow -1 < 3x+1 < 1 \Rightarrow -2 < 3x < 0 \Rightarrow -\frac{2}{3} < x < 0$$

Eg 1

$$y = x e^{1/x}$$

$$D: x \in \mathbb{R} \setminus \{0\}$$

$$\lim_{x \rightarrow -\infty} x e^0 = -\infty \cdot 1 = -\infty$$

$$\lim_{x \rightarrow +\infty} x e^0 = +\infty \cdot 1 = +\infty$$

No orizzontali,
cerchiamo oblique

$$\lim_{x \rightarrow \pm\infty} \frac{x e^x}{x} = e^0 = 1 \quad m=1$$

$$\lim_{x \rightarrow \pm\infty} x e^x - x = x (e^x - 1) = \frac{e^y - 1}{y} = 1 \quad q=1$$

$y = 1/x$
 $y \rightarrow 0$

L'asintoto obliquo c'è ed ha retta $y = x+1$

$$\lim_{x \rightarrow 0^-} x e^{1/x} = 0 \cdot e^{-\infty} = 0 \cdot 0 = 0$$

$$\lim_{x \rightarrow 0^+} x e^{1/x} = \frac{e^y}{y} = +\infty \quad \text{perché}$$

$y = 1/x$
 $y \rightarrow +\infty$
infiniti

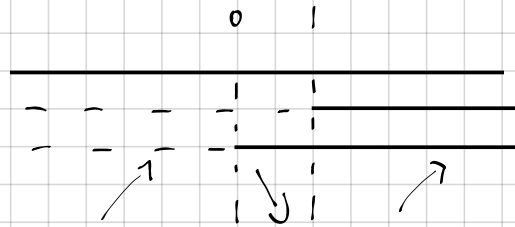
C'è un asintoto verticale a 0^+

$$f(x) = x e^{1/x}$$

$$f'(x) = (1)(e^{1/x}) + (x)(e^{1/x} \cdot -\frac{1}{x^2}) = e^{1/x} + (x e^{1/x}) \left(-\frac{1}{x^2}\right) = e^{1/x} + \left(-\frac{x e^{1/x}}{x^2}\right) = e^{1/x} - \frac{e^{1/x}}{x} = e^{1/x} \left(1 - \frac{1}{x}\right)$$

$$y' = e^{1/x} \left(1 - \frac{1}{x}\right)$$

$$\begin{cases} e^{1/x} > 0 \\ 1 - \frac{1}{x} > 0 \end{cases} \Leftrightarrow \begin{cases} \forall x \\ x - 1 > 0 \\ x > 0 \end{cases} \Leftrightarrow \begin{cases} \forall x \\ x > 1 \\ x > 0 \end{cases}$$



$f(1)$ è minimo locale

$$\lim_{h \rightarrow 0^-} \frac{f(h) - f(0)}{h} = \frac{h e^{1/h}}{h} = e^{1/h} = e^{-\infty} = 0$$

$$\lim_{h \rightarrow 0^+} \frac{f(h) - f(0)}{h} = \frac{-h e^{1/h}}{h} = -e^{1/h} = -e^{-\infty} = 0$$

f è derivabile in $x=0$ e $f'(0)=0$

Eg 2A

$$y = e^{x/2} - \sqrt{1+x}$$

Sono 2 funzioni $f(x)$ e $g(x)$ con $f(x) \neq 0$ e $\lim_{x \rightarrow x_0} f(x) = \lim_{x \rightarrow x_0} g(x) = 0$, $f(x)$ e $g(x)$ sono derivabili intorno a x_0 e

$$\lim_{x \rightarrow x_0} f'(x)/g'(x) = l \in \mathbb{R}, \text{ allora } \lim_{x \rightarrow x_0} f(x)/g(x) = l$$

$$\lim_{x \rightarrow 0} \frac{e^{x/2} - \sqrt{1+x}}{x^2} = \frac{1-1}{0} = \frac{0}{0}$$

$$f'(x) = \frac{e^{x/2}}{2} - \frac{1}{2\sqrt{1+x}}$$

$$g'(x) = 2x$$

$$f''(x) = \frac{1}{2} e^{x/2} + \frac{1}{4(1+x)^{3/2}}$$

$$g''(x) = 2$$

$$\lim_{x \rightarrow 0} \frac{\frac{e^{x/2}}{2} - \frac{1}{2\sqrt{1+x}}}{2x} = \frac{1/2 - 1/2}{0} = \frac{0}{0}$$

$$\frac{\frac{1}{2} e^{x/2} + \frac{1}{4(1+x)^{3/2}}}{2} = \frac{1/2 + 1/4}{2} = \frac{3/4}{2} = \frac{3}{8}$$

Eg 3A

$$f(x) = 4x \lg(x)$$

$$\int 4x \lg x = 2x^2 \lg x - \int 2x^2 \frac{1}{x} = 2x^2 \lg x - x^2 + C$$

$$F(1) = 2 \cdot 1 \cdot 0 - 1 + C = 2 \Rightarrow 0 - 1 + C = -2 \Rightarrow C = -1$$

$$2x^2 \lg x - x^2 - 1 = F(x)$$

Per far sì che una serie possa convergere $\lim_{n \rightarrow +\infty} a_n = 0$

$$\lim_{x \rightarrow 0} 2x^2 \lg x - x^2 - 1 = x^2 (2 \lg x - 1) - 1 = x^2 (2 \lg x - 1) - 1 = x^2 \lg x - x^2 - 1 = 0 - 0 - 1 = -1$$

La serie derivata da questa funzione non può convergere